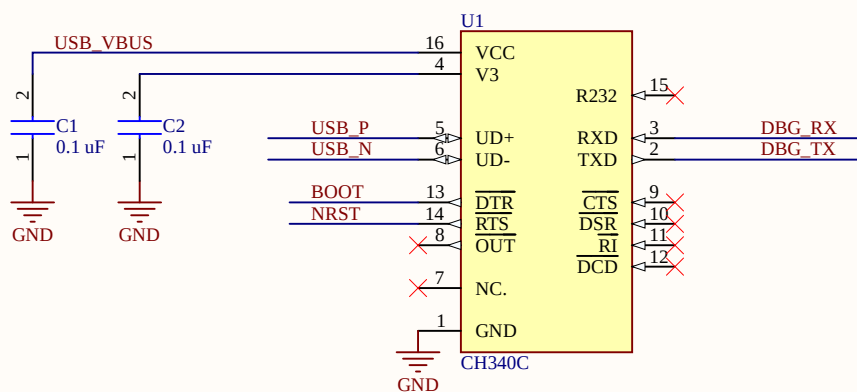
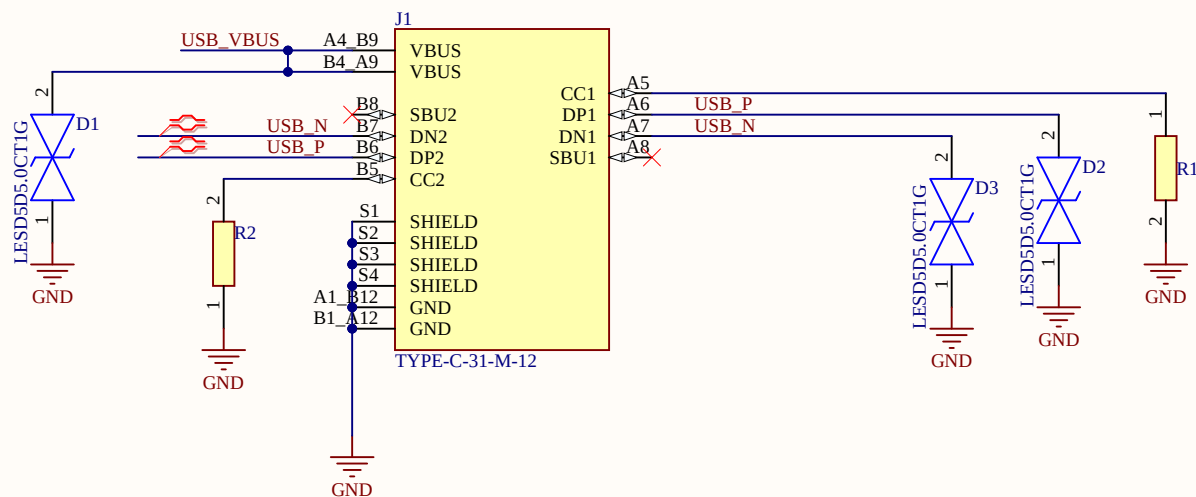




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File:	C:\Users\...\programmer.SchDoc	Drawn By:

- **Battery:** 3.6 V, typical 4 years life (15 s logging interval)
- **Battery Diagnostics:** Voltage, remaining capacity, low battery alarm
- **Operating Temperature:** -30 °C to +70 °C
- **Relative Humidity:** 95% non-condensing
- **Housing:** IP66 ABS or better
- **RTC:** Real-Time Clock, auto sync, daylight saving change
- **Certifications:** CE; IEC 61000-4-2, -4-3, -4-4, -4-5, -4-6, 61000-6-4

Communications

- **GPRS:** 4G (auto 3G/2G fallback), SMS, TCP/IP client & server
- **TCP Connections:** 4 total (2 inbound, 2 outbound)
- **Antenna:** 2.4 dBi internal (std.), 5 dBi external (opt.)
- **SIM:** Internal micro-SIM
- **Bluetooth:** v5.2 (2 Mbps)
- **Protocols:** Auto-detect Native, Modbus RTU/TCP Slave

Inputs

- **Impedance:** $\geq 10\text{ M}\Omega$
- **Overvoltage Protection:** $\pm 60\text{ V AC/DC}$
- **Voltage Range:** $\pm 20\text{ V AC/DC}$
- **Current Range:** $\pm 0.2\text{ A AC/DC}$
- **Current Density:** 0-400 A/m² (coupon area 5-25 cm²)
- **Measurement Time:** < 50 ms
- **Coupon Switching:** Solid-state relay

Storage Capacity

- On measurements: 30 days @ 15 s logging
- Eoff & Enat: 365 days @ 1 day logging
- Event log: last 250 events
- Alarm log: last 1000 alarms
- Config changes: last 450
- Battery history: 10 years (monthly)

https://documentation.espressif.com/esp32-wroom-32e_esp32-wroom-32ue_datasheet_en.pdf
https://documentation.espressif.com/esp32-wrover-e_esp32-wrover-le_datasheet_en.pdf

[LCSC link](#)

Pin	IO	Function
1	1	NRST
2	2	EN
3	3	EN
4	4	EN
5	5	EN
6	6	EN
7	7	EN
8	8	EN
9	9	EN
10	10	EN
11	11	EN
12	12	EN
13	13	EN
14	14	EN
15	15	EN
16	16	EN
17	17	EN
18	18	EN
19	19	EN
20	20	EN
21	21	EN
22	22	EN
23	23	EN
24	24	EN
25	25	EN
26	26	EN
27	27	EN
28	28	EN
29	29	EN
30	30	EN
31	31	EN
32	32	EN
33	33	EN
34	34	EN
35	35	EN
36	36	EN
37	37	EN
38	38	EN
39	39	EN
40	40	EN
41	41	EN
42	42	EN
43	43	EN
44	44	EN
45	45	EN
46	46	EN
47	47	EN
48	48	EN
49	49	EN
50	50	EN
51	51	EN
52	52	EN
53	53	EN
54	54	EN
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57	57	EN
58	58	EN
59	59	EN
60	60	EN
61	61	EN
62	62	EN
63	63	EN
64	64	EN
65	65	EN
66	66	EN
67	67	EN
68	68	EN
69	69	EN
70	70	EN
71	71	EN
72	72	EN
73	73	EN
74	74	EN
75	75	EN
76	76	EN
77	77	EN
78	78	EN
79	79	EN
80	80	EN
81	81	EN
82	82	EN
83	83	EN
84	84	EN
85	85	EN
86	86	EN
87	87	EN
88	88	EN
89	89	EN
90	90	EN
91	91	EN
92	92	EN
93	93	EN
94	94	EN
95	95	EN
96	96	EN
97	97	EN
98	98	EN
99	99	EN
100	100	EN

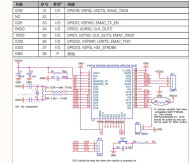
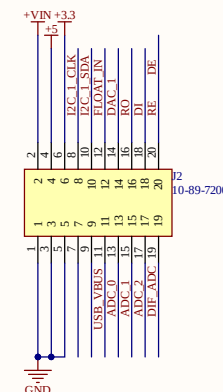
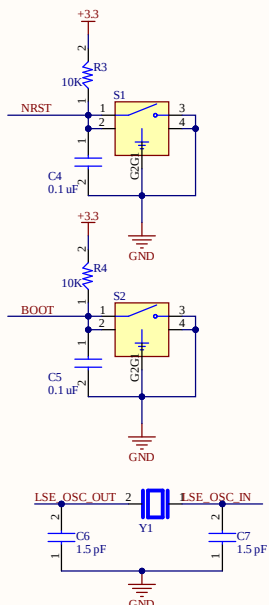
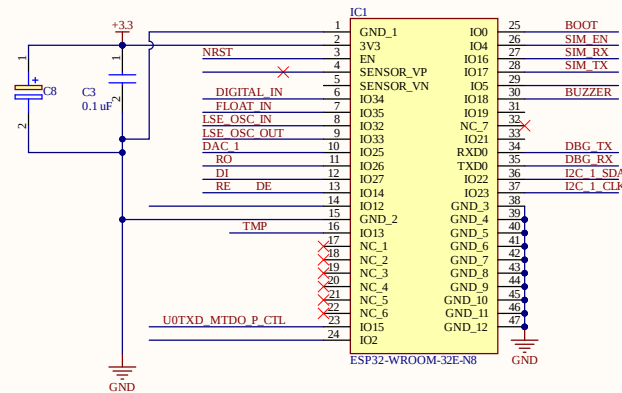


Table 4: Default Configuration of Strapping Pins

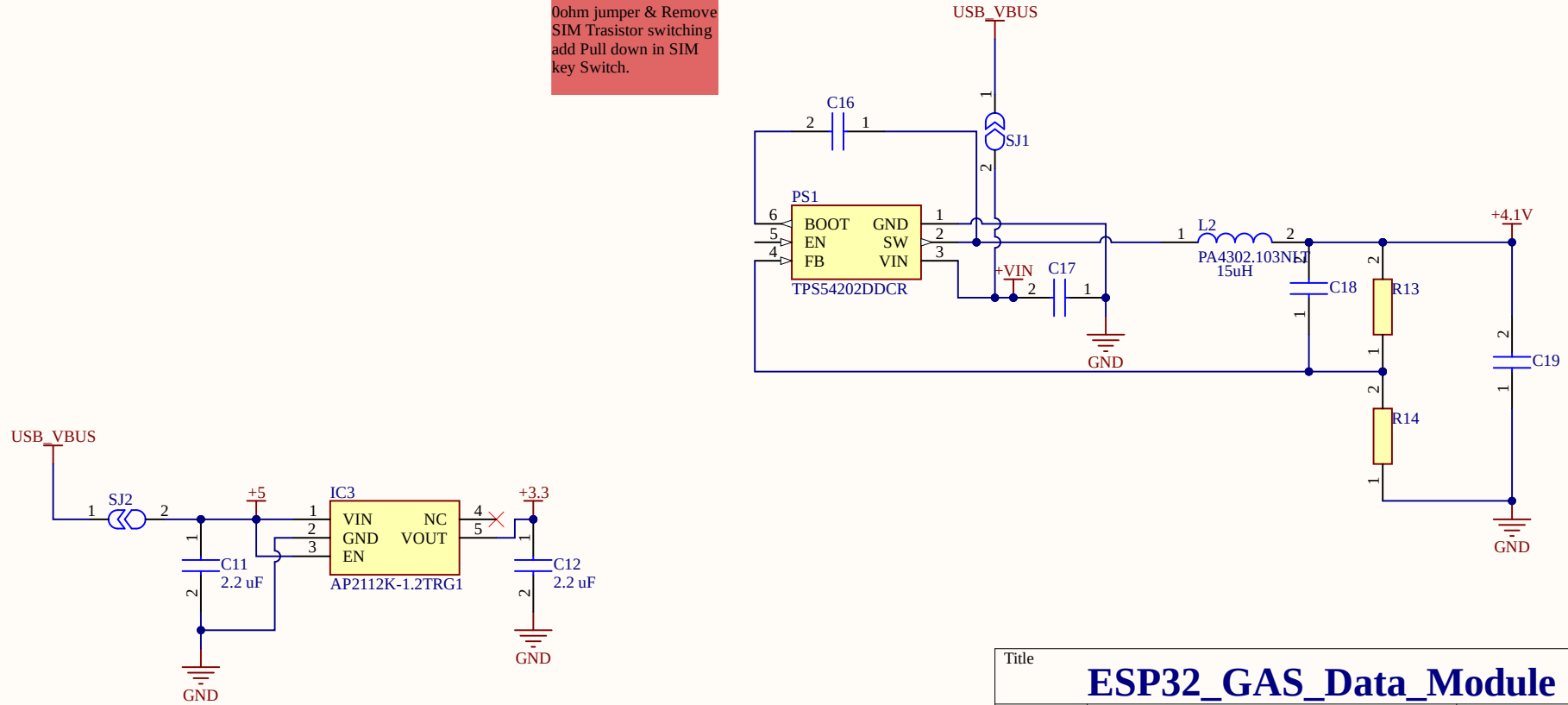
Strapping Pin	Default Configuration	Bit Value
GPIO0	Pull-up	1
GPIO2	Pull-down	0
MTDI	Pull-down	0
MTDO	Pull-up	1
GPIO5	Pull-up	1

Strapping Pin	Default Configuration	Bit Value
GPIO0	Pull-up	1
GPIO2	Pull-down	0
MTDI	Pull-down	0
MTDO	Pull-up	1
GPIO5	Pull-up	1

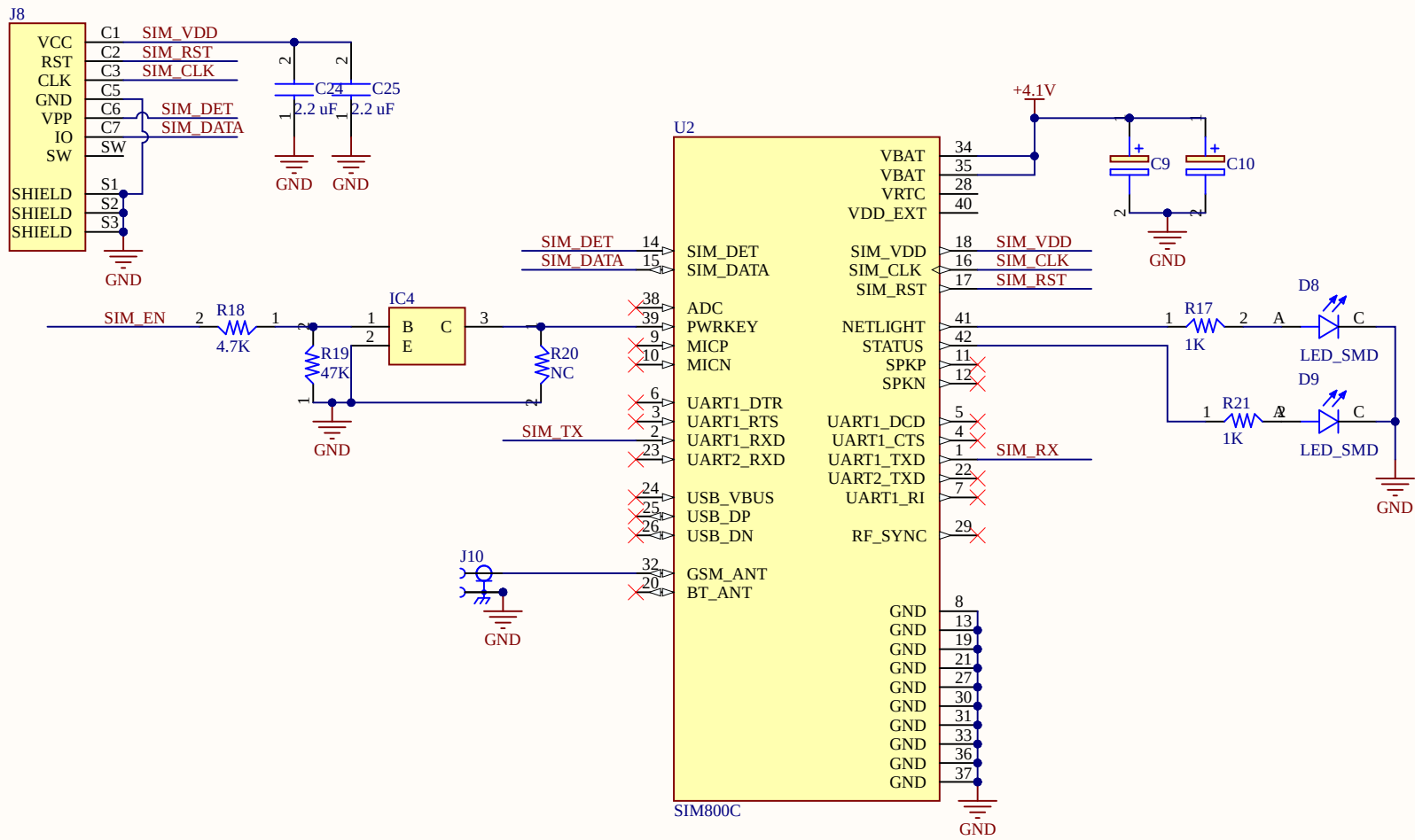


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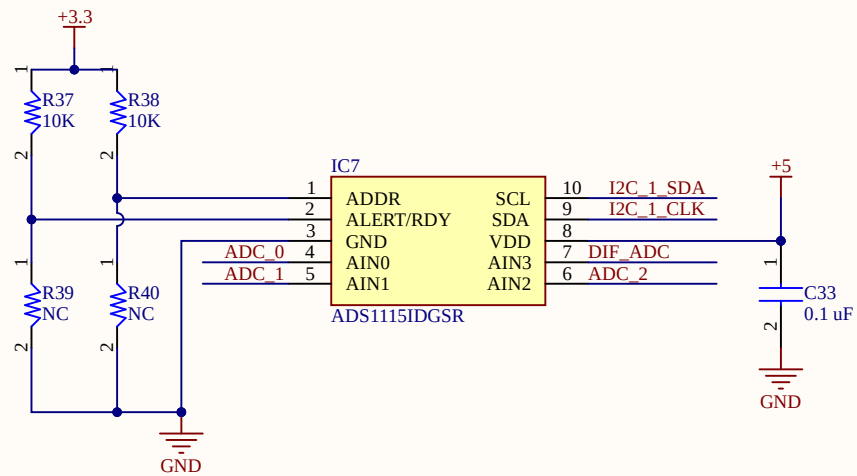
ADD EN pin for sim
Power On OFF With
0ohm jumper & Remove
SIM Trasistor switching
add Pull down in SIM
key Switch.



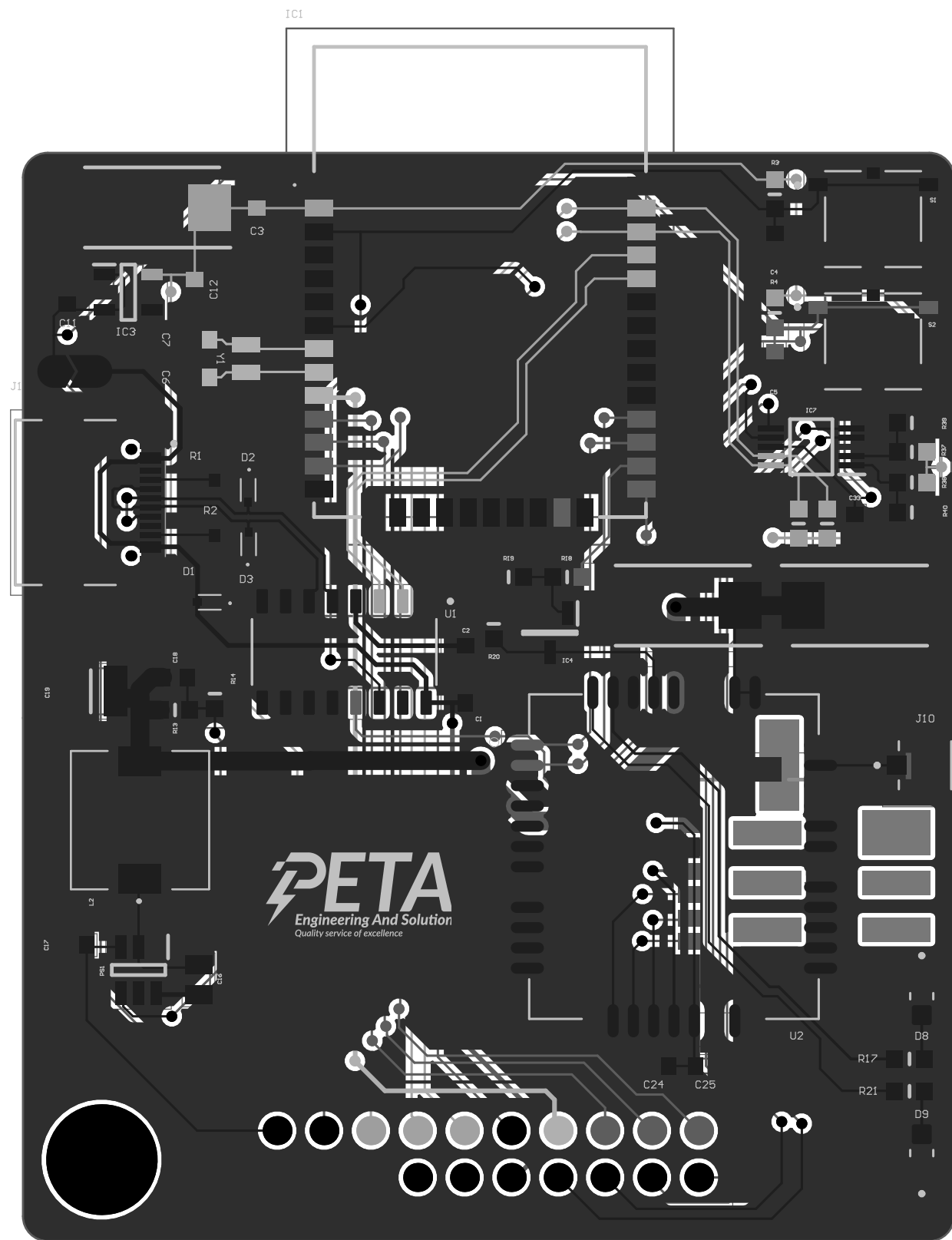
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Size	Number	Revision		
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Title		
Size	Number	Revision
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Title		
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Comment	Description	Designator	Footprint	LibRef	Quantity
0.1 uF	Capacitor	C1, C2, C3, C4, C5, C33	CAP0603	CC0603KRX7R9BB104	6
1.5 pF	Capacitor	C6, C7	CAP0603	CC0603KRX7R9BB104	2
TPSD227K010R0100	Capacitor Polarised	C8, C9, C10	CAPPM7344X310N	TPSD227K010R0100	3
2.2 uF	Capacitor	C11, C12, C24, C25	CAP0603	CC0603KRX7R9BB104	4
CD805C104M5RAC-TU	Capacitor	C16	CAPC2012X88N	CD805C104M5RAC-TU	1
C5750X5R1H106K230K A	Capacitor	C17	CAP0603	C5750X5R1H106K230K A	1
GRM0335C1E560JA01 D	Capacitor	C18	CAP0603	GRM0335C1E560JA01 D	1
GRM32ER61C476KE15 L	Capacitor	C19	CAPC3225X270N	GRM32ER61C476KE15 L	1
LES5D5.0CT1G	TVSDiode (Bi-directional)	D1, D2, D3	LES5D50CT1G	LES5D5.0CT1G	3
LED_SMD	Red 633nm LED Indication - Discrete 2.1V 0805 (2012 Metric)	D8, D9	LEDSC125X200X90-2N	5990110007F	2
ESP32-WROOM-32E-N8	Integrated Circuit	IC1	ESP32WROOM32EN8	ESP32-WROOM-32E-N8	1
AP2112K-1.2TRG1	Integrated Circuit	IC3	SOT95P285X140-5N	AP2112K-1.2TRG1	1
MMBT2222ALT1G	Integrated Circuit	IC4	SOT96P237X111-3N	MMBT2222ALT1G	1
ADSL115IDGSR	Integrated Circuit	IC7	SOP50P490X110-10N	ADSL115IDGSR	1
TYPE-C-31-M-12	USB Connectors 24 Receptacle 1 8.94* 7.3mm RoHS	J1	HRO_TYPEC-31-M-12	TYPE-C-31-M-12	1
10-89-7200	Connector	J2	HDRV20W87P254_2X10_2553X508X864P	10-89-7200	1
SIM8060-6-1-14-00-A	6 Position Card Connector NANO SIM Surface Mount, Right Angle Gold	J8	GCT_SIM8060-6-1-14-00-A	SIM8060-6-1-14-00-A	1
CONURL001-SMD		J10	UNIX_CONURL001-SMD	CONURL001-SMD	1
PA4302.103NLT	Inductor	L2	PA4302103NLT	PA4302.103NLT	1
TPS54202DDCR	Power Supply	PS1	SOT95P280X110-6N	TPS54202DDCR	1
CROW0402100KFKEDHP	Resistor	R1, R2	RESCL005X40N	CROW0402100KFKEDHP	2
10K	Resistor	R3, R4, R37, R38	RES0603	RES	4
4.7K	Resistor	R5, R6, R18	RES0603	RES	3
CROW0402100KFKEDHP	Resistor	R13	RES0603	CROW0402100KFKEDHP	1
CROW040216K9FKEDHP	Resistor	R14	RES0603	CROW040216K9FKEDHP	1
1K	Resistor	R17, R21	RES0603	RES	2
47K	Resistor	R19	RES0603	RES	1
NC	Resistor	R20, R39, R40	RES0603	RES	3
PTS526_SMG15_SMTR 2_LFS	Tactile Switch SPST-NO Top Actuated Surface Mount	S1, S2	SW_PTS526_SMG15_SMTR2_LFS	PTS526_SMG15_SMTR 2_LFS	2
SOLDER_JUMPER_1206		S11, S12	JUMP_SOLDER_JUMPER_1206	SOLDER_JUMPER_1206	2
CH340C	USB to serial chip CH340	U1	SOIC127P600X180-16N	CH340C	1
SIM800C	Module: GSM/Bluetooth; GPRS; 2G; 1800MHz; 900MHz; 85.6kbps; SMD	U2	LCC110P1570X1760X250-42N	SIM800C	1
CM8V-T1A-32.768KHZ-6PF-20PPM-TA-QC	Tuning Fork Crystal 32.768kHz in SMD 1.2 x 2.0mm package	Y1	XTAL_CM8V-T1A-32.768KHZ-6PF-20PPM-TA-QC	CM8V-T1A-32.768KHZ-6PF-20PPM-TA-QC	1